

REVIEW PROBLEMS

ANSWERS

Please note that this is NOT a practice exam – the length and style of questions aren't necessarily indicative of the length and style of the actual exam. This is just a review of problems addressing *some* of the concepts you learned so far.

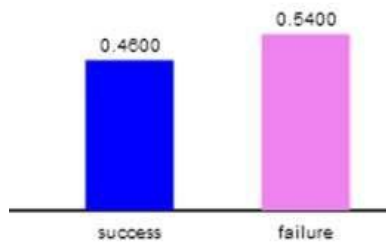
QUESTION 1

- a.
Categorical
- b.
Quantitative (if recorded as number of minutes between arrival and seen by doctor).
- c.
Categorical
- d.
This is a research question/comparison, not a variable posed to individual visitors (the variables asked of the visitors would be length of wait and gender)
- e.
This is a summary of the data, not a variable posed to individual visitors, the variable would be whether or not the patient arrived before noon
- f.
Categorical
- g.
Quantitative
- h.
Categorical
- i.
This is a summary of the data, not a variable posed to individual visitors (see f)
- j.
Categorical (Even though recorded values are numbers, ordinary arithmetic operations make no sense)

QUESTION 2

a.

A bar graph showing the proportion that landed up (success) and the proportion that landed down (failure):



b.

Statistic: $\hat{p} = \frac{46}{100} = 0.46$ (sample proportion of spins that landed with the logo facing up.) **Note:** You should get into the habit of showing all your work, and writing down all relevant quantities (like the numerator and denominator; don't jump straight to the decimal answer.)

c.

The parameter of interest is the long-run proportion of spins that the racquet would land up.

d.

Null hypothesis: the long-run proportion of spins that the racquet would land up is .50

Ho: $\pi_{\text{up}} = 0.5$, where π denotes the long-run proportion of spins that the racquet would land logo-up.

Alternative hypothesis: the long-run proportion of spins that the racquet would land up is not .50

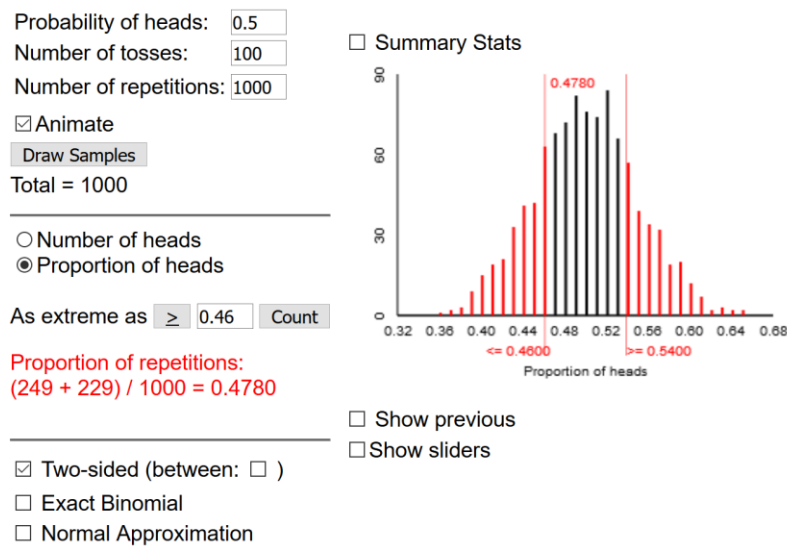
Ha: $\pi_{\text{up}} \neq 0.5$

We are using a two-sided alternative hypothesis because there is no clear “direction” in the research question. We wouldn't want to use the method of spinning the racquet to see who serves first if either the racquet lands up more than half the time or if it lands up less than half the time.

e.

Flip a coin 100 times and record the number of times the coin lands heads. Let "heads" represent the racquet landing “up.” Repeat this process a large number of times and look at the distribution of the proportion of heads across all these trials. See where 0.46 falls in this distribution.

f.



Yes, the analysis was conducted correctly.

- The “probability of heads” is 0.5 which is correct as this is the hypothesized value we stated under the null hypothesis for question 2d.
- The “number of tosses” is 100 which is the sample size of the study (100 spins of the racquet).
- The “number of repetitions” is also correct as it is a large number.
- The input next to “as extreme as” of 0.46 is also correct as this is the observed statistic of 0.46 (sample proportion of spins that landed with the label facing up).
- The “two-sided” test ration button was selected which is also correct since we are conducting a two sided test with the alternative hypothesis of $H_a: \pi_{up} \neq 0.5$.

g.

Based on the sample proportion of racquet spins in the sample that landed up of 0.46, we obtained a large p-value of 0.48. This p-value gives us weak evidence against the null hypothesis, so we fail to reject it. We do **not** have sufficient evidence that the long-run proportion of spins that the racquet would land up differs from 0.5. In other words, it is **plausible** to say that the long-run proportion of spins that the racquet would land up is 0.50.

QUESTION 3

a)

One teacher

b)

Whether the teacher is African American or not.

Levels: African American or not.

Alternative formulation: the variable is the race of the teacher, with levels “African American” or “Not African American.”

c)

$$15/405 = 0.0370$$

d)

The parameter of interest is the long-run proportion of teachers hired by the school district that are African American.

e)

$$\pi_{AA}$$

f)

$$H_0: \pi_{AA} = 0.154$$

$$H_a: \pi_{AA} < 0.154$$

g)

Describe process:

Probability of heads:

Number of tosses:

Number of repetitions:

☒ Show animation

Total Repetitions = 0

☐ Summary Statistics

Choose statistic:

☒ Number of heads

☐ Proportion of heads

Count samples

As extreme as

☐ Show previous results

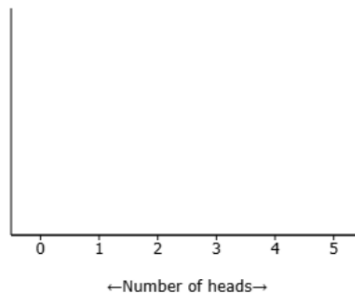
☐ Show sliders

Options:

☐ Two-sided

☐ Exact Binomial

☐ Normal Approximation



probability of heads = 0.154

number of tosses = 405

the number of repetitions is some larger number like 1,000 or 10,000

h)

$$0.154$$

i)

Describe process:

Probability of heads:

Number of tosses:

Number of repetitions:

☒ Show animation

Total Repetitions = 1000

Choose statistic:

☐ Number of heads

☒ Proportion of heads

Count samples

As extreme as

Options:

☐ Two-sided

☐ Exact Binomial

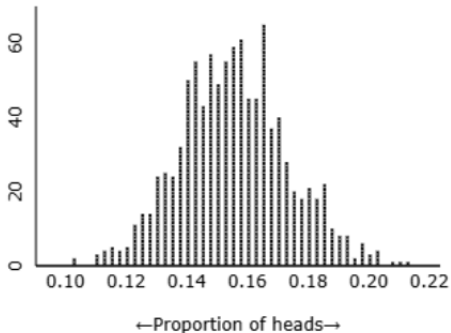
☐ Normal Approximation

Most recent results

Number of Heads = 69

Number of Tails = 336

☐ Summary Statistics



←Proportion of heads→

☐ Show previous results

☐ Show sliders

We should use the sign “ $\leq$ ” in the blue box and the value of 0.0370 in the red box.

j)

0.00

k)

Very strong evidence against the null hypothesis

l)

The p-value of 0.00 is the proportion of all simulated samples with a sample proportion equal to 0.037 or smaller, under the assumption that the null hypothesis is true ($\pi = 0.154$).

m)

Yes, there are 15 successes (African American) and 390 failures (not African American). Since both the successes and failures are greater than 10, we can state that the validity conditions are met.

n)

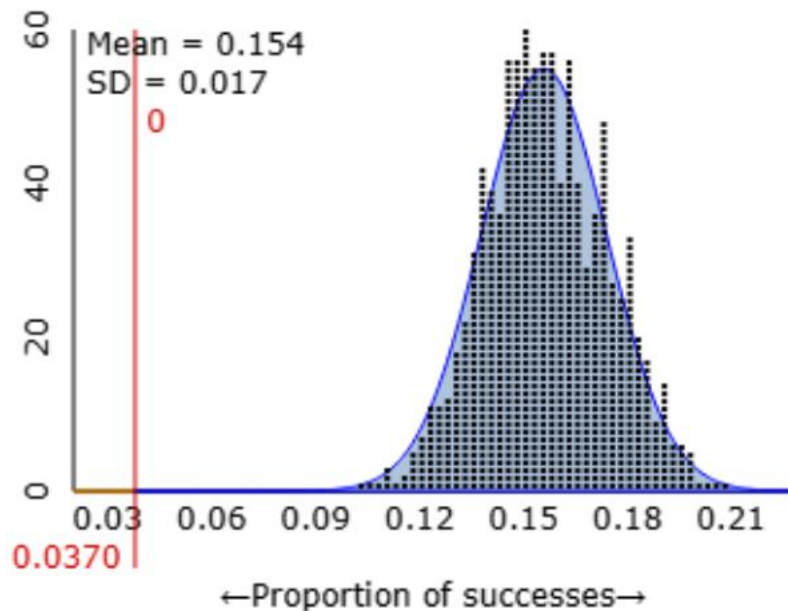
$$SD_{\text{NULL}} = \sqrt{(0.154 \cdot (1 - 0.154) / 405)} = 0.0179$$
$$z = (0.0370 - 0.154) / 0.0179 = -6.54$$

o)

The sample proportion of hired teachers who are African Americans of 0.154 is more than 6 standard deviations below the value of 0.154 (hypothesized long-run proportion).

p)

Yes, the normal distribution is a good approximation to the simulation results and we expected this to be the case because the validity conditions are met.



q)

The results agree, a large (absolute value) standardized statistic (e.g., larger than 2) will correspond to a small p-value (both the simulation-based and theory-based p-values are very very small).

r)

The new standardized statistic is $(0.037 - 0.057)/0.0179 = -1.12$

The standardized statistic (z) is closer to zero compared to the previous value and it suggests weaker evidence against the null hypothesis because now the observed sample proportion is not as far from the hypothesized long-run proportion of 0.154.

QUESTION 4

a

Skewed to right

b

0.6

c

150-180 seconds

d

420-450 seconds

e

Decrease

f

Decrease

QUESTION 5

Smallest: B Next-to-smallest: C Next-to-largest: D Largest: A

QUESTION 6

a

G

b

E

QUESTION 7

A person

QUESTION 8

a

Whether or not the person owns a pet bird

Whether or not the person has lung cancer

b

Both categorical

QUESTION 9

a

2

b

Categorical

C

Asian

D

Hispanic

QUESTION 10

No, this sampling method would be biased toward large words

QUESTION 11

a

Categorical variable

b

Not a variable

c

Quantitative variable

d

Not a variable

e

Categorical variable

QUESTION 12

Blood type